Supporting Information

Physics-Consistent Generative Inverse Design of Angle-Resolved Fourier-Operator Metasurfaces

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This supporting information expands the methodological details that are only summarized in the main manuscript. The presentation below is organized as a conventional supplementary note rather than an implementation guide, with emphasis on dataset construction, forward surrogate design, and the theoretical and architectural details of the conditional diffusion inverse model.

1 Dataset Preparation and Task Metrics

1.1 Freeform structure generation and physical labeling

The design space is a family of freeform binary metasurface layouts with spatial size 64×64, where the binary convention is $1 =$ material and $0 =$ air. Each structure is generated from a coarse random field, symmetrized under the constraint class $C_4 + \sigma_x$, and then smoothed, binarized, and morphologically regularized so that overly thin bridges, isolated islands, and very small voids are discouraged. This construction intentionally enlarges the accessible geometry family beyond a low-dimensional parametric library while preserving fabrication-oriented regularity.

The target fill ratio is not fixed at a single value; instead, the dataset spans five nominal fill-ratio levels,

$$\text{fill} \in \{0.4, 0.5, 0.6, 0.7, 0.8\}, \tag{1}$$

and the geometric generation process further perturbs the coarse-grid scale, smoothing strength, and minimum feature threshold. As a result, the final structure set contains both topological diversity and systematic variation in material occupancy.

Each structure is labeled by rigorous coupled-wave analysis under a fixed material platform consisting of a silicon patterned layer on a silica substrate, illuminated from air. The sampling grid is defined jointly over wavelength, incidence angle, and polarization-preserving transmission channels, so the design target is not a scalar device label but a discretized response tensor.

1.2 Response tensor organization

For each structure, the physical label is written as

$$\mathbf{R}(\lambda, \theta) = \begin{bmatrix} T_{pp}(\lambda, \theta) \\ T_{ss}(\lambda, \theta) \end{bmatrix} \in \mathbb{R}^{2 \times 11 \times 17}. \tag{2}$$

The learning framework therefore operates on two coupled objects: a binary geometry $x \in \{0,1\}^{64 \times 64}$ and a discretized response tensor $\mathbf{R}$. During training, the response tensor is standardized pointwise:

$$\widetilde{\mathbf{R}}_{qjk} = \frac{\mathbf{R}_{qjk} - \mu_{qjk}}{\sigma_{qjk} + \varepsilon}, \tag{3}$$

Table 1: Dataset and labeling configuration.

Item	Configuration
Structure representation	Freeform binary pattern, 64×64 pixels
Symmetry prior	$C_4 + \sigma_x$
Binary convention	$1 =$ material, $0 =$ air
Nominal fill-ratio levels	$0.4, 0.5, 0.6, 0.7, 0.8$
Periodicity	500 nm
Patterned-layer thickness	500 nm
Structured material	Si
Input / output media	Air / SiO_2
Wavelength sampling	800–1300 nm, step 50 nm (11 points)
Angle sampling	$-40°$ to $40°$, step $5°$ (17 points)
Retained channels	T_{pp} and T_{ss} magnitudes
Response tensor size	$2 \times 11 \times 17$

where q indexes polarization channel, j indexes wavelength, and k indexes angle. This location-wise normalization preserves physically meaningful weak-response regions, such as operator zeros and polarization leakage bands, which would otherwise be suppressed by a single global scaling factor.

1.3 Task-specific metric family

The framework uses a family of task-aware scores rather than a single universal metric. For one-dimensional operator targets, the ideal angular envelope is written as

$$g_m(\theta) = \left| \frac{\sin\theta}{\sin\theta_{\max}} \right|^m, \tag{4}$$

where $m = 2, 3, 4$ denotes second-, third-, and fourth-order operator behavior. At each wavelength row, the score is decomposed into center suppression, shape agreement, edge support, and finite-bandwidth persistence:

$$S_m = (1 - w_b)\left(w_c S_c + w_s S_s + w_e S_e\right) + w_b S_b. \tag{5}$$

In the present study, the coefficient set is

$$w_c = 0.6, \qquad w_s = 0.3, \qquad w_e = 0.1, \qquad w_b = 0.2. \tag{6}$$

For polarization-independent operation, the two channels must simultaneously retain the desired row behavior while remaining matched at large angles:

$$S_{\mathrm{PI}} = (1 - \beta)\min(S_p, S_s) + \beta S_{\mathrm{match}}, \tag{7}$$

$$S_{\mathrm{match}} = 1 - \frac{|T_{pp}(+\theta_a) - T_{ss}(+\theta_a)| + |T_{pp}(-\theta_a) - T_{ss}(-\theta_a)|}{2A_{\max}}, \tag{8}$$

with $\beta = 0.15$ and $\theta_a = 40°$ in the present evaluation protocol.

For polarization-multiplexed operation, one channel is designated as active while the other is suppressed:

$$S_{\mathrm{sil}} = \max\left(0, 1 - \frac{R_{\mathrm{passive,max}}}{\tau}\right), \tag{9}$$

$$S_{\mathrm{PM}} = \frac{w_a S_{\mathrm{act}} + w_q S_{\mathrm{sil}}}{w_a + w_q}, \tag{10}$$

with $w_a = 0.7$ and $w_q = 0.3$.

For low-pass targets, the ideal row is modeled as

$$g_\sigma(\theta) = \exp\left(-\frac{\theta^2}{\sigma^2}\right), \tag{11}$$

and the current protocol searches over $\sigma \in \{8^\circ, 12^\circ, 16^\circ\}$ to identify the best-matching effective passband width. The ranking itself is driven mainly by the row-shape term, while center transmission and edge rejection are retained as diagnostic quantities.

For spatiotemporal differentiation, the target is evaluated on a two-dimensional wavelength–angle window:

$$G^\star(\lambda, \theta) \propto k_x^2(\theta)\big(\omega(\lambda) - \omega_0\big)^2. \tag{12}$$

The corresponding score combines zero-line suppression, corner-lobe preservation, and auxiliary agreement terms:

$$S_{\mathrm{ST}} = 0.50 S_{\mathrm{zero}} + 0.40 S_{\mathrm{corner}} + 0.10 S_{\mathrm{aux}}, \tag{13}$$

where the auxiliary contribution balances weighted template error and modal projection consistency.

Table 2: Task-specific metrics used in the present framework.

Task	Target form	Main score components	Physical emphasis
Second / third / fourth order	Row envelope $g_m(\theta)$	center, shape, edge, bandwidth	operator zero, angular growth, outer-angle support, finite spectral persistence
Polarization-independent	matched row response in both channels	per-channel row score + large-angle matching	simultaneous operator behavior and polarization consistency
Polarization-multiplexed	active row + silent passive row	active-shape + passive suppression	one channel performs the task while the other remains quiet
Low-pass	Gaussian row envelope $g_\sigma(\theta)$	shape-dominant score with center / edge diagnostics	central passband and outer-angle rejection
Spatiotemporal differentiation	2D window $k_x^2(\omega - \omega_0)^2$	zero lines, corners, weighted error, projection	joint wavelength–angle morphology rather than a single row

Figure Placeholder

Suggested final content: freeform structure generation, RCWA labeling on the wavelength–angle grid, and a compact schematic of the row-wise and 2D scoring logic.

Figure 1: Placeholder for the dataset preparation and metric definition overview. A final figure can integrate the geometric generation process, the response-grid labeling stage, and representative score decompositions for row and window targets.

2 Forward Surrogate Architecture and Design Rationale

2.1 Forward mapping

The forward surrogate approximates the map

$$\mathcal{F}_\phi : \{0,1\}^{64\times 64} \to \mathbb{R}^{2\times 11\times 17}, \tag{14}$$

which predicts the joint T_{pp} and T_{ss} magnitudes over the full wavelength–angle sampling grid. Its role is deliberately limited but physically important: it provides a fast approximation to the structure-to-response map for candidate screening and supplies the response-consistency term used later in diffusion training.

2.2 Architecture summary

The surrogate is constructed as a coordinate-aware convolutional encoder followed by multiscale response-grid fusion. Two Cartesian coordinate channels are concatenated to the binary geometry, allowing the network to correlate planar layout with directional scattering. A hierarchy of residual blocks then extracts features at progressively coarser scales. The deepest latent representation is globally pooled, passed through a multilayer perceptron, and broadcast back onto the response grid. Finally, all feature levels are interpolated to the 11×17 response resolution and fused together with the explicit response-grid coordinates.

This architecture can be summarized as

$$\widehat{\mathbf{R}} = H(P_0, P_1, P_2, P_3, G_{\text{global}}, C_{\lambda\theta}), \tag{15}$$

where P_ℓ denotes projected multiscale encoder features, G_{global} denotes the broadcast latent summary, and $C_{\lambda\theta}$ denotes the response-grid coordinate field.

2.3 Design rationale

The coordinate channels are useful because angle-resolved scattering is not a translation-invariant texture problem. The multiscale encoder is needed because operator performance depends on both small subwavelength boundary features and large-scale topology. The global broadcast branch allows every wavelength–angle location to access long-range structural statistics such as fill ratio, connectivity, and gross symmetry. Joint prediction of T_{pp} and T_{ss} is preferable to training two unrelated models because both channels originate from the same geometry and their correlation structure is part of the task definition itself.

2.4 Training protocol

The surrogate is trained on the standardized response tensor using an L_1 loss,

$$\mathcal{L}_{\text{fwd}} = \left\| \mathcal{F}_\phi(x) - \widetilde{\mathbf{R}} \right\|_1 . \tag{16}$$

The present training setup uses a batch size of 64, AdamW optimization with learning rate 10^{-4} and weight decay 10^{-4}, a 90%/10% train–validation split, gradient clipping at 1.0, ReduceLROnPlateau learning-rate scheduling, and early stopping. Symmetry-preserving flips are used as data augmentation during training.

Table 3: Detailed architecture of the forward surrogate.

Stage	Resolution / channels	Main operation	Architectural role
Input assembly	64×64, $1+2$ channels	binary structure plus (x, y) coordinates	breaks pure translation invariance and exposes geometric position information
Stem block	64×64, 32 channels	convolution + residual refinement	extracts low-level boundary and occupancy features
Encoder stage 1	32×32, 64 channels	residual downsampling	captures short-range morphology with enlarged receptive field
Encoder stage 2	16×16, 128 channels	residual downsampling	fuses mesoscale topology and local motifs
Encoder stage 3	8×8, 256 channels	residual downsampling	aggregates global structure-aware context
Latent refinement	8×8, 256 channels	stacked residual blocks	stabilizes the deepest representation before fusion
Global context branch	pooled latent $\rightarrow$ 128 channels	adaptive pooling + MLP + broadcast	injects device-level topology and effective fill information into every response location
Multiscale projection	all stages $\rightarrow 11 \times 17$	bilinear interpolation to response grid	aligns all learned features with the physical wavelength–angle manifold
Grid-coordinate injection	11×17, 2 channels	explicit response-plane coordinates	distinguishes central / outer angles and short / long wavelengths
Fusion block	11×17, concatenated multiscale tensor	convolutional fusion across scales	combines local, global, and coordinate information on the output grid
Prediction head	11×17, 2 channels	final convolutional regression head	jointly predicts T_{pp} and T_{ss} maps

Table 4: Forward surrogate training protocol.

Item	Setting
Loss	pointwise normalized-space L_1 regression
Optimizer	AdamW
Learning rate / weight decay	10^{-4} / 10^{-4}
Batch size	64
Train–validation split	90% / 10%
Stabilization	gradient clipping, adaptive learning-rate decay, early stopping
Augmentation	symmetry-preserving flips only
Validation readout	denormalized physical-space MAE in addition to normalized loss

Figure Placeholder

Suggested final content: input structure plus coordinate channels, residual encoder stages, global latent broadcast, multiscale projection to the response grid, and the dual-channel output head.

Figure 2: Placeholder for the forward surrogate architecture. A final figure can present the coordinate-aware encoder, the multiscale response-grid fusion pathway, and the dual-polarization prediction head in one consolidated diagram.

3 Conditional Diffusion Inverse Design: Theory, Architecture, and Conditioning

3.1 Forward diffusion process

The inverse problem is treated as a conditional diffusion process because a single target response may correspond to multiple admissible structures. Let $x_0 \in [-1, 1]^{1\times64\times64}$ denote a clean structure image after linear remapping from the binary interval $[0, 1]$. The forward noising Markov chain is written as

$$q(x_t|x_{t-1}) = \mathcal{N}\Big(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I\Big)\,, \tag{17}$$

with a cosine schedule for β_t. By repeated composition, the marginal distribution at step t has the closed form

$$q(x_t|x_0) = \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t)I\big)\,, \tag{18}$$

where

$$\alpha_t = 1-\beta_t, \qquad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s. \tag{19}$$

Equivalently,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{20}$$

3.2 Reverse model and velocity parameterization

The reverse process seeks a parameterized model

$$p_\theta(x_{t-1}|x_t, \mathbf{c}), \tag{21}$$

conditioned on the target response tensor $\mathbf{c}$. In standard diffusion formulations, one may predict either the injected noise ϵ, the clean sample x_0, or a linear combination of the two. In the present framework, the chosen target is the velocity parameter

$$v^\star = \sqrt{\bar{\alpha}_t}\, \epsilon - \sqrt{1-\bar{\alpha}_t}\, x_0. \tag{22}$$

This choice is convenient because the pair $(x_t, v^\star)$ can be linearly inverted to recover both ϵ and x_0. Starting from

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \tag{23}$$

$$v^\star = \sqrt{\bar{\alpha}_t}\, \epsilon - \sqrt{1-\bar{\alpha}_t}\, x_0, \tag{24}$$

one directly obtains

$$\hat{x}_0 = \sqrt{\bar{\alpha}_t}\, x_t - \sqrt{1-\bar{\alpha}_t}\, \hat{v}, \tag{25}$$

$$\hat{\epsilon} = \sqrt{1-\bar{\alpha}_t}\, x_t + \sqrt{\bar{\alpha}_t}\, \hat{v}, \tag{26}$$

where $\hat{v} = v_\theta(x_t, t, \mathbf{c})$ is the network prediction. This is the parameterization used throughout training and sampling.

3.3 Training objective

The diffusion loss is the mean-squared error between predicted and target velocity:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0,\epsilon,t}\left[\|v_\theta(x_t, t, \mathbf{c}) - v^\star\|_2^2\right]. \tag{27}$$

This basic objective is then augmented with two physically motivated regularizers:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{bin}}\mathcal{L}_{\text{bin}}. \tag{28}$$

The response-consistency term is

$$\mathcal{L}_{\text{phys}} = \left\| \mathcal{F}_{\phi}\left(\frac{\hat{x}_0 + 1}{2}\right) - \mathbf{c} \right\|_1, \tag{29}$$

which compares the surrogate-predicted response of the denoised sample against the target condition. The binarization term is

$$\mathcal{L}_{\text{bin}} = \text{mean}(x(1-x)), \tag{30}$$

which penalizes ambiguous gray values in the denoised prediction. In the present training setup,

$$\lambda_{\text{phys}} = 0.5, \qquad \lambda_{\text{bin}} = 0.05. \tag{31}$$

3.4 Conditional representation and conditioning mechanisms

The target response tensor

$$\mathbf{c} \in \mathbb{R}^{2\times 11\times 17} \tag{32}$$

is encoded by a dedicated condition encoder rather than by a single label embedding. The condition encoder outputs three complementary objects:

1. a global condition embedding,
2. multiscale condition feature maps,
3. a sequence of spectral tokens.

These three condition objects enter the U-Net in distinct ways. First, multiscale condition maps are additively injected into residual blocks after suitable resizing. Second, the global condition embedding is concatenated with the time embedding and converted into scale–shift modulation coefficients inside residual blocks. Third, token-level cross-attention is applied in the deeper latent hierarchy, allowing spatial features to attend directly to the full spectral condition sequence.

3.5 Classifier-free guidance and sampling

Classifier-free guidance is implemented by randomly dropping the true condition during training and replacing it with a learned null condition. At inference time, the conditional and unconditional predictions are combined as

$$v_{\text{cfg}} = v_{\text{uncond}} + s_{\text{cfg}}\left(v_{\text{cond}} - v_{\text{uncond}}\right), \tag{33}$$

where the default guidance strength is $s_{\text{cfg}} = 3.0$. Sampling starts from Gaussian noise, iterates the reverse process from $t = T-1$ to 0, remaps the final result from $[-1,1]$ to $[0,1]$, and thresholds it at 0.5 to obtain a binary candidate structure.

Table 5: Detailed architecture of the conditional diffusion inverse model.

Module	Resolution / form	Main operation	Architectural role
Noisy input branch	64×64, 1 channel	diffusion-state input convolution	initializes the spatial denoising stream
Time embedding path	vector embedding	sinusoidal embedding + MLP	encodes diffusion time step for all residual blocks
Condition encoder stem	11×17, multi-channel	coordinate-aware convolutional encoding	extracts structured response features from the target tensor
Condition downsampling path	$11 \times 17 \to 6 \times 9 \to 3 \times 5$	residual downsampling	builds multiscale condition maps aligned with U-Net depths
Global condition embedding	vector embedding	pooled condition summary	provides compact response-level context for scale–shift modulation
Spectral token projection	token sequence	flatten + linear projection	converts deep condition features into tokens for cross-attention
Encoder residual stage 1	shallow spatial level	residual block + condition-map injection	couples early geometry denoising to high-resolution condition maps
Encoder residual stage 2	intermediate spatial level	residual block + condition-map injection	matches mesoscale geometry updates to intermediate condition maps
Deep encoder stage	deep spatial level	residual block + self-attention + cross-attention	lets latent geometry features access global condition tokens
Bottleneck block	deepest latent level	residual blocks + self-attention + cross-attention	performs globally conditioned denoising at the most compressed representation
Decoder stage 1	deep upsampling level	skip fusion + residual block + cross-attention	reconstructs geometry while retaining token-level global conditioning
Decoder stage 2	intermediate upsampling level	skip fusion + residual block	restores mid-scale structure with condition-map support
Decoder stage 3	shallow upsampling level	skip fusion + residual block	reconstructs high-resolution binary boundaries under local conditioning
Output projection	64×64, 1 channel	normalization + final convolution	predicts velocity field on the structure plane
Classifier-free guidance	conditional / unconditional pair	guided score combination during sampling	strengthens target adherence in reverse diffusion
Physics-consistency branch	surrogate-evaluated denoised sample	response-consistency regularization	ties generation to the target electromagnetic response
Binarization regularizer	denoised sample in $[0, 1]$	gray-value penalty	encourages sharp binary outputs

Figure Placeholder

Suggested final content: the forward diffusion process, the velocity-parameterized reverse model, multiscale condition injection, cross-attention, and classifier-free guidance.

Figure 3: Placeholder for the conditional diffusion formulation. A final figure can combine the forward and reverse diffusion equations with the conditional U-Net, explicitly marking multiscale condition injection, global modulation, and token-level cross-attention.